

Homework 2. Model selection and regularization: A model for the average fare

70-467 - Machine Learning for Business Analytics

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Background. You have average fares for 638 routes in the United States. A route is a flight from one city to another. Each row is one route. It is not one ticket. These routes have already been flown. FARE is the average price paid on that route, in dollars. In this dataset it runs from about \$42 to about \$402.

The dataset also records how far apart the two cities are, whether passengers usually change planes, whether Southwest flies the route, how the traffic is split among carriers, the income and population of each city, and whether the airports limit takeoffs or gates. City names and airport codes identify the route. The codebook on the last page gives the type and the unit of every variable.

Airlines use information like this when they set a fare. A long route, a route where passengers change planes, or a route flown by only one or two carriers does not get the same fare as a short nonstop route with several carriers.

The problem. We want a predictive model for the average fare on a route, built from the other routes on this dataset. There are many variables, so there are many possible models. If you put every variable into one least-squares fit, or you keep a variable because it looks important, the fit on the routes you used can look better and better. That does not mean the model will predict well on a route you did not use.

Please use the starter. Answer each question in the notebook, in the place where it asks.

The video. Imagine that you are recording a presentation of your analysis for the Vice President of Pricing and Revenue Management. The Vice President may consider your analysis when deciding what average fare to count on in the revenue plan, how far off that plan may be, and which features of the route that fare depends on. The revenue plan is built on that fare. Use the answers you have already written.

Before you record, answer these for yourself. Then use what you decided in the presentation.

1. What decision will this person make with your estimate?
2. What does the Vice President need to hear in order to use it, and what can stay in the notebook?
3. Which model are you handing the Vice President, and which variables does it use? Why did the others stay out?
4. How will you show the size of the error, in dollars, so the Vice President can tell whether to count on the estimate? What will you set that error next to?
5. Where two models are close, why are you using the one you chose? Does shrinking the coefficients change that reason?
6. Where could this person misuse your estimate?
7. Which graph carries that case? What does its title say, in dollars? If you show another graph, can you say in a sentence or two how it bears on the choice?
8. How will you order the presentation so the decision is clear before the methods?

Deliverables.

Submit on Canvas by Sunday, 4 October 2026, 11:59 pm:

- a knitted R notebook (HTML or PDF from the qmd), with the code you add and your written answers,
- at most six slides, and
- a video of at most five minutes.

The notebook, the slides, and the video must use the same model and the same dollar figures.

The camera must be on. I need to see you delivering. Please refrain from reading the slides. That is a zero on the slides and the video. AI can assist you in preparing, but it should not be used to deliver the video.

Please start early and give this your full effort. The homework is not only a great tool to help you apply and solidify the concepts covered so far in sessions 3 and 4, but also a good preparation for the midterm on Wednesday, 7 October. Please reach out if you need support.

Codebook. 638 rows, 18 columns. One row = one route in the United States, from one city to another. FARE is the outcome.

Variable	Type	Unit	Description
S_CODE	text	airport code	Origin airport.
S_CITY	text	city	Origin city.
E_CODE	text	airport code	Destination airport.
E_CITY	text	city	Destination city.
COUPON	numeric	coupons	Average number of flight coupons. One coupon is one flight.
NEW	numeric	carriers	Carriers that began flying this city pair.
VACATION	text	Yes / No	Yes if the city pair is coded as a vacation route.
SW	text	Yes / No	Yes if Southwest flies this city pair.
HI	numeric	index	Concentration of passengers among carriers, up to 10,000.
S_INCOME	numeric	dollars	Average personal income in the origin city.
E_INCOME	numeric	dollars	Average personal income in the destination city.
S_POP	numeric	people	Population of the origin city.
E_POP	numeric	people	Population of the destination city.
SLOT	text	Free / Controlled	Controlled: an airport at one end limits takeoffs and landings.
GATE	text	Free / Constrained	Constrained: an airport at one end has few gates left.
DISTANCE	numeric	miles	Miles between the two cities.
PAX	numeric	passengers	Passengers recorded on this city pair.
FARE	numeric	dollars	Average price paid. This is the outcome.